

TrES-5b

R-0944

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_230618 · created 2026-07-29T13:19:18+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460113.8146 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.054 +/- 0.038
Transit depth (Rp/Rs)^2	0.29 +/- 0.41 [%]
Orbital Inclination (inc)	87.21 +/- 3.0
Airmass coefficient 1 (a1)	0.98 +/- 0.02
Airmass coefficient 2 (a2)	0.019 +/- 0.024
Scatter in the residuals of the lightcurve fit is	2.09 %
Best Comparison Star	#2 - [529, 461]
Optimal Method	PSF photometry
Transit Duration (day)	0.072 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.054 $\pm$ 0.038 vs 0.142 (deviation 2.3 σ)
Duration fit vs published	0.072 d vs 0.0758 d (deviation 0.27 σ)
Mid-time O-C	3.4 min
Depth SNR	1.4
Residual scatter	2.09 %

Light curve

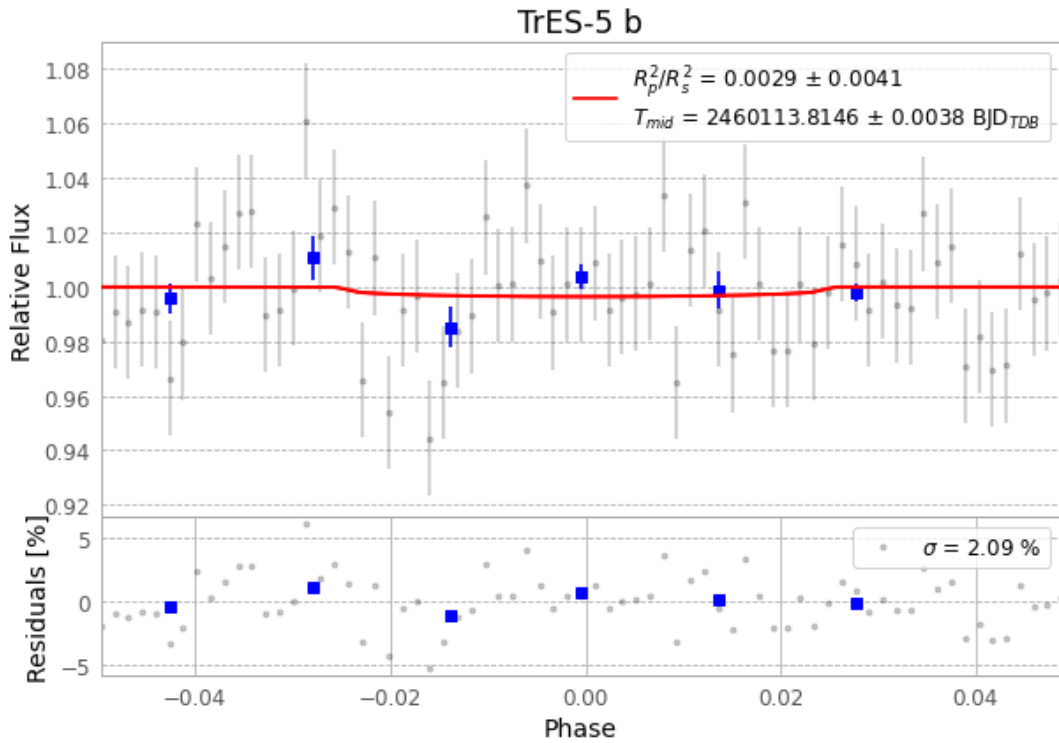


Figure 1 — FinalLightCurve_TrES-5 b_2023-06-17.png (SHA-256 a975656c0e5e1017...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 221], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f3a03e954ed317d9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), TRES-5230618054515.FITS → TRES-5230618091815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-230618.log (ac7df0f32c83...), FinalLightCurve_TrES-5 b_2023-06-17.png (a975656c0e5e...), FinalLightCurve_TrES-5 b_2023-06-17.csv (1ee60a8a57af...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 13:19Z.